Solution:

1. A. 7.5 lb

$$N = W = 150 \text{ lb}$$
$$F = \mu N = (0.05)(150) = 7.5 \text{ lb}$$
$$F_{forward} = F = 7.5 \text{ lb} \rightarrow \text{Ans}$$

Reference: GEAS 04-01 – Types of Equilibrium

2. B. 160 kN

$$N = W = 400 \text{ kN}$$
$$F = \mu N = (0.40)(400) = 160 \text{ kN}$$
$$P = F = 160 \text{ kN} \rightarrow \text{Ans}$$

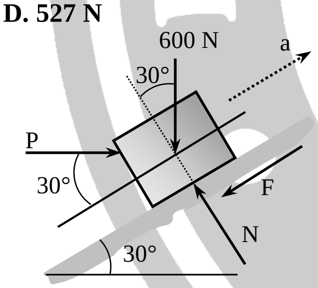
Reference: GEAS 04-01 – Types of Equilibrium

3. C. 1.33

$$\bar{y} = \frac{2a+b}{a+b}\frac{h}{3} = \frac{2(2)+4}{2+4}\frac{3}{3} = 1.33$$

Reference: GEAS 04-06 – Center of Mass

4. D. 527 N



$$\sum F_x = 0$$
$$P\cos 30° = F + 600\sin 30°$$
$$P\cos 30° = \mu N + 600\sin 30° \rightarrow \text{Eq. 1}$$

$$\sum F_y = 0$$
$$N = 600\cos 30° + P\sin 30° \rightarrow \text{Eq. 2}$$

Substitute Eq. 2 in Eq. 1:

$$P\cos 30 = 0.2\left[600\cos 30 + P\sin 30\right] + 600\sin 30$$
$$0.866P = 103.923 + 0.1P + 300$$
$$P = 527.31 \text{ N} \rightarrow \text{Ans}$$

Reference: GEAS 04-01 – Types of Equilibrium

5. A. 12

$$I = \frac{3mr^2}{10} = \frac{3(10)(2)^2}{10} = 12 \rightarrow \text{Ans}$$

6. A. 0.054

$$N = W = 65 \text{ lb}$$
$$F_{applied} = F = 1200 \text{ lb}$$
$$F = \mu N$$
$$1200 = \mu(65)$$
$$\mu = 0.054 \rightarrow \text{Ans}$$

Reference: GEAS 04-01 – Types of Equilibrium

7. A. $0.152W \le P \le 6.59W$

Use the formula for belt friction. Note that angle of contact should be in radians:

$$\theta = 1.5(2\pi) = 3\pi \text{ rad}$$
$$T_2 = T_1 e^{\mu\theta} = T_1 e^{0.2(3\pi)} = 6.59T_1$$

Therefore,

$$0.152W \le P \le 6.59W$$

Reference: GEAS 04-05 – Belt Friction

8. C. 250 N

Solve for the maximum static friction (friction that should be overcome before the box can move):

$$\sum F_y = 0$$
$$N - W = 0$$
$$N = W = mg$$
$$f_{s(max)} = \mu_s N$$
$$f_{s(max)} = (0.4)(100\text{kg})\left(9.81\frac{m}{s^2}\right)$$
$$f_{s(max)} = 392.4\text{N}$$

Since the force applied (250 N) did not reach the maximum static friction, the box will not yet move. In this case, the static friction is also 250 N.

Reference: GEAS 04-04 – Static and Kinetic Friction

9. B. N=1909 N, P=47 N

$$\Sigma F_x = 0$$
$$P\cos(\theta_1 + \theta_2) - \mu_s N + Mg\sin(\theta_2) = 0$$
$$\Sigma F_y = 0$$
$$N - Mg\cos\theta_2 + P\sin(\theta_1 + \theta_2) = 0$$
$$N = 1909 \text{ N}$$
$$P = 47 \text{ N}$$

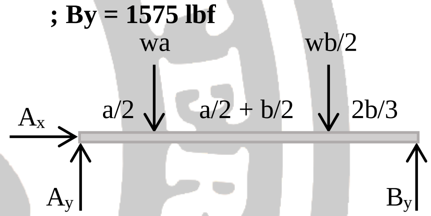
Reference: GEAS 04-01 – Types of Equilibrium

10. C. 30 lb

$$\Sigma F_y = 0; \quad N - W = 0$$
$$N = 100 \text{ lb}$$
$$\Sigma F_x = 0; \quad P - F = 0$$
$$P = \mu_s(N) = 30 \text{ lb}$$

Reference: GEAS 04-01 – Types of Equilibrium

11. Correction: Ax = 0 ; Ay = 2475 lbf ; By = 1575 lbf



$$\Sigma F_x = 0; A_x = 0 \rightarrow \text{Ans}$$
$$\Sigma M_A = 0;$$
$$-wa\frac{a}{2} - \frac{1}{2}wb\left(a + \frac{b}{3}\right) + B_y(a+b) = 0$$
$$B_y = 7 \text{ kN} \rightarrow \text{Ans}$$
$$\Sigma F_y = 0$$
$$A_y + B_y - wa - \frac{1}{2}wb = 0$$
$$A_y == 11 \text{ kN} \rightarrow \text{Ans}$$

Convert kN to lb (or lbf).
1 N = 0.225 lbf

Reference: GEAS 04-03 – Beams

12. A. 30.4 lb

$$\Sigma F_y = 0$$
$$N - W - F\sin\theta = 0$$
$$\Sigma F_x = 0$$
$$F\cos\theta - \mu_s N = 0$$
$$F = 30.4 \text{ lb}$$

Reference: GEAS 04-01 – Types of Equilibrium

13. A. (1.5, -5)

Use the formula to get the center of mass:

$$\bar{x} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$\bar{y} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

Note that the center of mass of the 15 kg bar is at the origin if you draw a cartesian plane:

$$\bar{x} = \frac{(5)(-9)+(10)(9)+(15)(0)}{5+10+15} = 1.5$$

$$\bar{y} = \frac{(5)(-6)+(10)(-12)+(15)(0)}{5+10+15} = -5$$

Reference: GEAS 04-06 – Center of Mass

14. D. 2 kg

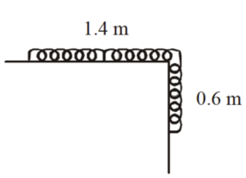
Taking the x-axis to be parallel to the plane and the y-axis perpendicular to the plane, and assuming that the system is still in equilibrium (static friction is used):

$$\sum F_y = 0$$
$$N - W_y = 0$$
$$N = mg\cos\theta$$
$$\sum F_x = 0$$
$$W_x - f_s = 0$$
$$mg\sin\theta - f_s = 0$$
$$mg\sin\theta = f_s$$
$$m(9.81)\sin 30 = 10$$
$$m = 2\text{kg} \rightarrow \text{Ans}$$

Reference: GEAS 04-01 – Types of Equilibrium

15. A. 3.6 J

The center of mass of the hanging part is at 0.3 m from the table.



$$m_{hanging} = \frac{4\text{x}0.6}{2} = 1.2 \text{ kg}$$

$$W = mgh = (1.2)(9.81)(0.3)$$
$$W = 3.6\text{J} \rightarrow \text{Ans}$$

Reference: GEAS 04-01 – Types of Equilibrium

16. C. 0.375

Figure a

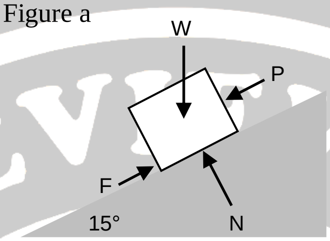
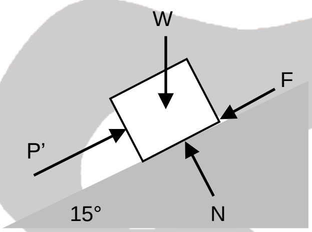


Figure b



For Figure a:

$$\Sigma F_y = 0$$
$$N - W_y = 0$$
$$N = W\cos 15°$$
$$\Sigma F_x = 0$$
$$F - P - W_x = 0$$
$$F = P + W\sin 15°$$
$$P = \mu_s W\cos 15° - W\sin 15°$$

For Figure b:

$$\Sigma F_y = 0$$
$$N - W_y = 0$$
$$N = W\cos 15°$$
$$\Sigma F_x = 0$$
$$P' - F - W_x = 0$$
$$P' = \mu_s W\cos 15° + W\sin 15°$$

Relate P and P'

$$P' = 6P$$
$$m_s W\cos 15° + W\sin 15° = 6(m_s W\cos 15° - W\sin 15°)$$
$$m_s \cos 15° + \sin 15° = 6(m_s \cos 15° - \sin 15°)$$
$$m_s = 0.375 \rightarrow \text{Ans}$$

Reference: GEAS 04-04 – Static and Kinetic Friction

17. C. Law of Acceleration

Take note that "rate" of change in momentum refers to how fast the momentum is changing. That means, it's the change in momentum per unit time.

$$F = \frac{\Delta mv}{t} = \frac{m\Delta v}{t} = ma$$

Reference: GEAS 04-01 – Types of Equilibrium

18. A. the same at ever point on its line of action

Principle of Transmissibility: The state of rest or of motion of a rigid body is unaltered if a force acting on the body is replaced by another force of the same magnitude and direction but acting anywhere on the body along the line of action of the applied forces.

Reference: GEAS 04-01 – Types of Equilibrium

19. A. algebraic sum of their moments about any point is equal to the moment of their resultant force about the same point

Varignon's Theorem: The theorem that states that the torque of a resultant of two concurrent forces about any point is equal to the algebraic sum of the torques of its components about the same point.

Reference: GEAS 04-01 – Types of Equilibrium

20. Is dependent on the area of contact, between the two surfaces

Static or Kinetic friction is always independent of the area of contact.

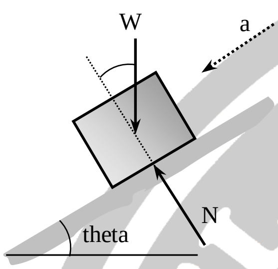
Reference: GEAS 04-04 – Static and Kinetic Friction

21. C. If the three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two

Lami's Theorem: It states that if three forces acting at a point are in equilibrium, each force is proportional to the sine of the angle between the two other forces.

Reference: GEAS 04-02 – Lami's Theorem

22. A. gsinθ



$$\sum F_x = ma$$
$$mg\sin\theta = ma$$
$$a = g\sin\theta$$

Reference: GEAS 04-01 – Types of Equilibrium

23. C. 3

A cantilever is a beam with a fixed support at one end. There are 3 reactions in a fixed support.

Reference: GEAS 04-03 - Beams

24. B. wL/2

For a uniformly varying load, the equivalent concentrated load is equal to the area of the figure. In the case of a uniformly varying load, the figure is a triangle.

Reference: GEAS 04-03 - Beams

25. D. Upward at its upper end

Since the ladder is leaning against a wall, the tendency is (1) for the upper end to slide down the wall and (2) the lower end to slide horizontally away from the wall. Since there is no friction on the ground, the friction present will be purely between the contact of the upper end of the ladder and the wall. Since the upper end has a tendency to slide downwards, friction must be pointing upwards. Friction is always opposite the direction of the motion or impending motion.

Reference: GEAS 04-04 – Static and Kinetic Friction

26. B. The box will not move

Solve for the maximum static friction (friction that should be overcome before the box can move):

$$\sum F_y = 0$$
$$N - W = 0$$
$$N = W = mg$$
$$f_{s(max)} = \mu_s N$$
$$f_{s(max)} = (0.4)(100kg)\left(9.81\frac{m}{s^2}\right)$$
$$f_{s(max)} = 392.4N$$

Since the force applied (350N) did not reach the maximum static friction, the box will not yet move. In other words, the force applied is not yet enough for the box to start moving.

Reference: GEAS 04-04 – Static and Kinetic Friction

27. C. The box will slide across the floor with an acceleration of 1.06 m/s^2

Based from the previous solution, the box needs to be pushed by at least 392.4 N for it to start moving. Since you are now applying a 400 N force to the box, the box will now be moving. Since the box is now moving, the friction present must be kinetic, and not static, friction.

$$\sum F_x = ma$$
$$P - f_k = ma$$
$$400 - \mu_k N = ma$$
$$400 - 0.3(100kg)\left(9.81\frac{m}{s^2}\right) = 100a$$
$$a = 1.06\frac{m}{s^2} \rightarrow Ans$$

Reference: GEAS 04-04 – Static and Kinetic Friction

28. C. independent on

Friction is always independent of the area of contact. If the area of contact increases or decreases, the magnitude and direction of friction will not change.

Reference: GEAS 04-04 – Static and Kinetic Friction

29. C. force of the floor on box 1 is greater

If you push only 1 box:

$$\sum F_y = 0$$
$$N - W = 0$$
$$N = W$$
$$\sum F_x = 0$$
$$P - f = 0$$
$$P = f = \mu N = \mu mg$$

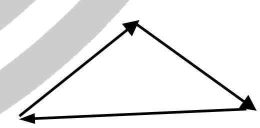
If you push only 2 boxes with the same mass (one of top of the other):

$$\sum F_y = 0$$
$$N - 2W = 0$$
$$N = 2W = 2mg$$
$$\sum F_x = 0$$
$$P' - f = 0$$
$$P' = f = \mu N = \mu 2mg$$
$$P' = 2P$$

P' is greater than P because the normal force of the floor on the boxes is greater in the second scenario.

Reference: GEAS 04-01 – Types of Equilibrium

30. D



Using the tail to head method, the resultant force or net force is the force created by connecting the tail of the first vector to the head of the final vector once you connect all of the vectors together. Look for the option where the resultant force is 0. In the case, that is option D.

Reference: GEAS 04-01 – Types of Equilibrium

31. C. 0.5 cot θ

If the wall is smooth, it means that there is no friction. The friction present is only between the foot of the ladder and the floor.
Since the ladder is in equilibrium:

$$\sum F_y = 0$$
$$N_f = W = mg$$
$$\sum F_x = 0$$
$$N_w = f_f = \mu N_f = \mu mg$$

Let x be the distance of the base of the ladder to the wall and y be the distance of the upper tip of the ladder to the floor.
Get the summation of moments about the point where the base of the ladder is in contact with the floor:

$$\sum M_{base} = 0$$
$$W(0.5x) = N_w y$$
$$mg(0.5x) = \mu mgy$$
$$0.5x = \mu y$$
$$\mu = 0.5\frac{x}{y} = 0.5\cot\theta$$

Reference: GEAS 04-01 – Types of Equilibrium

32. A. 1000 N

Get the summation of moments about point A to eliminate the unknown reactions at A:

$$\sum M_A = 0$$
$$-T(4) + 800(3) = 0$$
$$T = 600N$$

Consider all forces acting on the beam (in equilibrium):

$$\sum F_x = 0$$
$$-600 + R_x = 0$$
$$R_x = 600N$$

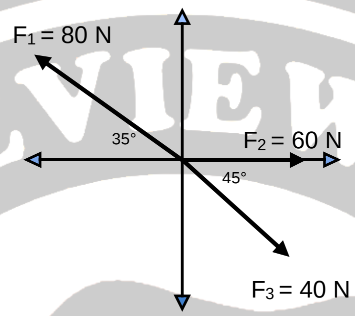
$$\sum F_x = 0$$
$$-800 + R_y = 0$$
$$R_y = 800N$$

Get the total reaction at the hinge (or get the vector sum of Rx and Ry):

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{600^2 + 800^2}$$
$$R = 1000N$$

Reference: GEAS 04-03 - Beams

33. A. 8.77 N at 217.7 degrees from +x



To make the system in equilibrium, the vector sum of all forces must be zero: $\sum F = 0$
Use your calculator:

$$\sum F = 0$$
$$0 = F_1 + F_2 + F_3 + F_4$$
$$F_4 = 0 - (F_1 + F_2 + F_3)$$
$$F_4 = 0 - (80\angle 145° + 60\angle 0 + 40\angle 315)$$
$$F_4 = 28.8\angle 217.7° \rightarrow \text{Ans}$$

Reference: GEAS 04-01 – Types of Equilibrium

34. D. 2.5 kg

Get the summation of vertical forces for Block B:

$$\sum F_x = 0$$
$$T - W_B = 0$$
$$T = W_B = m_B g$$

The tension in Block B should be the same as the tension in Block A.
Get the summation of forces in Block A:

$$\sum F_y = 0$$
$$N = W_A \cos 35°$$
$$N = m_A g \cos 35°$$

$$\sum F_x = 0$$
$$T + f = W_A \sin 35°$$
$$W_B + \mu N = W_A \sin 35°$$
$$m_B g + 0.4 m_A g \cos 35° = m_A g \sin 35°$$
$$m_B + 0.4(10)\cos 35° = 10 \sin 35°$$
$$m_B = 2.5kg$$

Reference: GEAS 04-01 – Types of Equilibrium

35. A. 51.3°

If the wall is smooth, it means that there is no friction. The friction present is only between the foot of the ladder and the floor.
Since the ladder is in equilibrium:

$$\sum F_y = 0$$
$$N_f = W = Mg$$
$$\sum F_x = 0$$
$$N_w = f_f = \mu N_f$$
$$N_w = (0.40)Mg$$

Let x be the distance of the base of the ladder to the wall and y be the distance of the upper tip of the ladder to the floor.

Get the summation of moments about the point where the base of the ladder is in contact with the floor:

$$\sum M_{base} = 0$$
$$W(0.5x) = N_w y$$
$$Mg(0.5x) = \mu Mgy$$
$$0.5x = \mu y$$
$$0.4 = 0.5\frac{x}{y} = 0.5\cot\theta$$
$$\theta = 51.3°$$

Reference: GEAS 04-01 – Types of Equilibrium

36. Correction: $\mu = 0.577$, d = 0.318 m

A block slides with constant velocity (acceleration = 0) down the plane:

$$\sum F_y = 0$$
$$N = mg\cos 30°$$

$$\sum F_x = 0$$
$$mg\sin 30° = \mu mg\cos 30°$$
$$\sin 30° = \mu\cos 30°$$
$$\mu = 0.577$$

The block is then projected up the plane with an initial speed of 2.5 m/s. Solve for distance travelled up the plane when it has stopped. Use Conservation of Energy:

$$KE_1 + PE_1 = KE_2 + PE_2 + loss$$
$$KE_1 + \cancel{PE_1} = \cancel{KE_2} + PE_2 + loss$$
$$\frac{1}{2}mv_1^2 = mgh + fd$$
$$\frac{1}{2}mv_1^2 = mgd\sin 30° + fd$$
$$\frac{1}{2}mv_1^2 = mgd\sin 30° + \mu mg\cos 30° d$$
$$\frac{1}{2}(2.5)^2 = (9.81)d\sin 30° + (0.577)(9.81)\cos 30° d$$
$$d = 0.318m$$

Reference: GEAS 04-04 – Static and Kinetic Friction

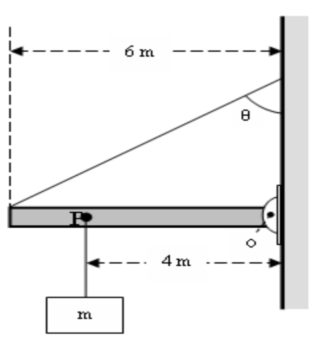
37. B. 200 N

Solution:
Get the summation of moments at the hinge:

$$\sum M = 0$$
$$+W(6) - (T\sin 30°)(6) = 0$$
$$+(200)(6) - (T\sin 30°)(6) = 0$$
$$T = 200N$$

Reference: GEAS 04-03 - Beams

38. A. 2 kN



Get the summation of moments at the hinge (forces at the hinge and the horizontal component of tension are not included in the equation):

$$\sum M = 0$$
$$-W(4) - W(3) + T_y(6) = 0$$
$$-W(4) - W(3) + (T\sin 30°)(6) = 0$$
$$-(80)(9.81)(4) - (100)(9.81)(3) + (T\sin 30°)(6) = 0$$
$$2027 = T$$

Reference: GEAS 04-03 - Beams

39. C. 24.4 N

Since there is no acceleration, take the summation of forces in both the x- and y-axis and equate to zero. Take the axis parallel to the plane as the x-axis and the axis perpendicular to the plane as the y-axis. Draw your FBD. From the FBD:

$$\sum F_y = 0$$
$$N = W_y = mg\cos 30°$$
$$\sum F_x = 0$$
$$P - W_x - f = 0$$
$$P = W_x + f$$
$$P = mg\sin 30 + 0.86mg\cos 30$$
$$P = 24.4N \rightarrow Ans$$

Reference: GEAS 04-01 – Types of Equilibrium

40. B. Dynamic Friction

The friction on a moving object is called kinetic or dynamic friction.
Reference: GEAS 04-04 – Static and Kinetic Friction

41. D. 120 degrees

Assume that the magnitude of the force is 5 N. Use your calculator and trial and error (plug in the choices in the red part):

$$F_1 + F_2 = R$$
$$5\angle 0° + 5\angle 120° = R$$
$$R = 5\angle 60°$$

It turns out that if the angle between F1 and F2 is 120 degrees, the resultant will have a magnitude that is the same as F1 and F2.
Reference: GEAS 04-01 – Types of Equilibrium

42. C. greater than

Static friction is always greater than the kinetic friction.
Reference: GEAS 04-04 – Static and Kinetic Friction

43. C. Limiting friction

Limiting friction is the maximum static friction on an object or is the friction experienced when the object is on the verge of motion.
Reference: GEAS 04-04 – Static and Kinetic Friction

44. B. Principle of resolution of forces
Reference: GEAS 04-01- Types of Equilibrium

45. A. 0 and 180 degrees

The resultant is maximum when the two forces complete adds up (that is the angle between them is 0).
The resultant is minimum when the two forces are in opposite direction (that is the angle between them is 180)
Reference: GEAS 04-01- Types of Equilibrium

46. C. Circular
Reference: GEAS 04-01- Types of Equilibrium

47. B. Q = R
Reference: GEAS 04-01- Types of Equilibrium

48. D. All of the above
Reference: GEAS 04-04 – Static and Kinetic Friction

49. A. Angle of friction
Reference: GEAS 04-04 – Static and Kinetic Friction

50. C. Greater than 1
Since static friction is greater, then the ratio of the static friction to dynamic friction must always be greater than 1.

Reference: GEAS 04-04 – Static and Kinetic Friction